Module 08: Planning in Elementary School

Learning Objectives

1. Define **Planning** within the context of Elementary School.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Elementary School to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Elementary School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Elementary School, we see Planning manifest in: * **Specific Example 1**: If you have 12 grapes and want to share them equally with 3 friends, you can plan ahead by counting out groups of 3 grapes each -- thinking about the answer before you start handing them out helps you make sure everyone gets the same amount! * **Specific Example 2**: When you want to build a block tower that is 5 blocks tall but you only have 3 blocks, you can make a plan: count how many more you need (5 minus 3 equals 2), then go find 2 more blocks before you start building so your tower turns out just right.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.